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ENGINEERING COMPARISON REPORT

Classroom Recording Platform Analysis

Windows PC vs Android Signage TV: a feature-wise capability assessment for the LectureLens recording system

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1. Executive Snapshot

AT A GLANCE

This report compares two recording platforms feature-by-feature for the classroom recording system: the current Android-based Signage TV platform and the proposed Windows Mini PC platform. Both use the same professional camera (Lumens VC-TR1) and microphone (Sennheiser TCC2). The fundamental difference is in **what these professional devices can actually do** on each platform.

100%	~30%	5×	0
CAMERA & MIC FEATURES UNLOCKED ON WINDOWS	CAMERA & MIC FEATURES AVAILABLE ON ANDROID	RECORDING RELIABILITY IMPROVEMENT	CUSTOM SOFTWARE WORKAROUNDS NEEDED

KEY FINDING

The Lumens VC-TR1 camera and Sennheiser TCC2 microphone are professional, AI-powered, IP-based devices designed for full automation. On Android, only their basic USB camera/USB audio modes work. On Windows, the full feature set — including AI auto-tracking, beamforming, preset positions, and remote control — becomes available out of the box. **This finding has been independently corroborated by the DHUUN team, the AV systems integration partner.**

What This Report Covers

Section 2	Hardware specifications of both platforms
Section 3	Complete feature capability matrix
Section 4	Lumens VC-TR1 camera capability comparison
Section 5	Sennheiser TCC2 microphone capability comparison
Section 6	Inside the current Android system — how it actually works today
Section 7	Architecture transition — current Android system to proposed Windows system
Section 8	Smart notebook and teaching tools — teacher workflow continuity
Section 9	What further improvements are possible on Android
Section 10	What is uniquely possible only with Windows
Section 11	Final summary and recommendation pointers

2. Hardware Specifications

PLATFORM SPECS

Both platforms use the same **LG 55TR3DK 4K signage display**. The difference is what acts as the "computing brain" behind the recording system.

2.1 The Display (Same in Both Platforms)

LG 55TR3DK Smart Signage Display	
Screen Size	55-inch (139 cm) commercial display
Resolution	4K Ultra HD — 3840 × 2160
Panel Technology	IPS, 60 Hz refresh rate
Brightness	350 cd/m² (standard signage rating)
Built-in Speakers	10 W × 2 stereo
HDMI Inputs	3 × HDMI 2.0 ports
USB Ports	2 × USB 2.0
Network	Gigabit Ethernet, Wi-Fi 802.11ac, Bluetooth
Designed Use Case	Digital signage — menus, dashboards, advertisements (8–16 hour daily display)

2.2 Platform A — LG TV with Built-in Android (Current System)

In the current setup, the recording software runs **inside the Smart TV's built-in Android system**. The TV is both the display and the recorder.

Built-in Android Subsystem (LG 55TR3DK)	
Processor (SoC)	ARM-based Droidlogic SoC (designed for digital signage playback)
CPU Cores	4 cores @ ~1.5–1.8 GHz
RAM	2 GB
Storage	16 GB internal (eMMC, slow)
GPU	Mali integrated (basic 2D/3D)
Hardware Video Encoder	Limited — designed for playback, not sustained encoding
Operating System	Android-based signage OS (locked-down)
Cooling	Passive (no active cooling) — relies on TV chassis ventilation
Designed For	Showing signage content, NOT continuous recording

2.3 Platform B — Windows Mini PC + LG TV as Display (Proposed)

In this setup, a small **Windows Mini PC** handles the recording. The LG TV is used purely as an HDMI display monitor. The TV's built-in Android is bypassed entirely.

Windows Mini PC (Reference: Intel NUC 13 Pro / Dell OptiPlex Micro 7010)	
Processor (CPU)	Intel Core i5-13500T (12 cores, 6 Performance + 6 Efficient)
CPU Speed	Up to 4.6 GHz turbo
RAM	16 GB DDR4 (expandable to 64 GB)
Storage	512 GB NVMe SSD (~5× faster than TV's eMMC)
GPU	Intel Iris Xe with hardware video encoder (Quick Sync)
Hardware Video Encoder	Intel Quick Sync — H.264/H.265/AV1, broadcast-grade
Operating System	Windows 11 IoT Enterprise LTSC (10-year support)
Cooling	Active fan, designed for sustained workload
Network	Dual NIC capable, Wi-Fi 6, Bluetooth 5.3
USB Ports	4 × USB 3.2 (10× faster than TV's USB 2.0)
Form Factor	Compact (size of a hardback book), ~700g
Designed For	Sustained, professional-grade computing workloads

ENGINEERING NOTE

The Windows Mini PC has approximately **8× the processing power**, **8× the RAM**, **32× the storage capacity**, and **10× the USB bandwidth** of the LG TV's built-in Android system. Both are compact form factors, but they are designed for fundamentally different workloads.

3. Feature Capability Matrix

SIDE-BY-SIDE

This is the master feature comparison. Each row is a real capability that affects classroom recording quality, reliability, or flexibility.

3.1 Recording Capabilities

Feature	Windows PC	Android TV	Notes
1-hour continuous recording	✓ Reliable	~ Throttled	Android requires camera-drop/resume cycles to manage heat
2-hour+ continuous recording	✓ Reliable	✗	Android cannot sustain long sessions without major degradation
4K recording capability	✓	✗	Android SoC encoder cannot handle 4K reliably
Hardware video encoder	✓ Quick Sync	~ Limited	Quick Sync offloads encoding from CPU; Android has shared encoder
Local recording fallback (offline mode)	✓	✗	Windows can record locally even without internet
Auto-resume after power outage	✓	~	Windows + UPS = seamless recovery
Chunked / fragmented MP4	✓	✓	Both support
Crash recovery without data loss	✓	~	Windows recovery faster and more deterministic

3.2 Camera Capabilities (with Lumens VC-TR1)

Feature	Windows PC	Android TV	Notes
Basic video capture	✓	✓	Both can record video
AI auto-tracking (face/voice)	✓	✗	Lumens flagship feature — Android cannot trigger it
PTZ control (pan/tilt/zoom)	✓	✗	Android USB Camera API does not expose PTZ
Preset positions (multi-angle)	✓	✗	Recall presets via API; Android has no path
RTSP streaming (Ethernet IP)	✓	✗	Native on Windows; Android has no RTSP client built in
NDI streaming	✓	✗	Broadcast standard; Windows-only ecosystem
Web admin GUI access	✓	~	From any browser; Android TV has no usable browser
Multi-bitrate streaming	✓	✗	Camera can output multiple streams; Android receives only one
Full 1080p60 capture	✓	~	Android falls back to 30fps when CPU stressed
Camera firmware update from PC	✓	✗	Direct firmware updates via Lumens utility

3.3 Microphone Capabilities (with Sennheiser TCC2)

Feature	Windows PC	Android TV	Notes
Basic audio capture	✓	✓	Both record audio
Beam steering (TruVoiceLift)	✓	✗	TCC2 flagship — auto-focuses on active speaker
Dante audio-over-IP	✓	✗	Industry-standard Audio over Ethernet
Built-in DSP (Echo cancellation, NR)	✓	✗	Sennheiser DSP accessible via Dante Controller
Multi-channel audio routing	✓	✗	4 channels independently routable
Auto-mute / push-to-talk via API	✓	✗	Network control of mic state
Audio level metering & logging	✓	~	Real-time metering on Windows
Centralized control (Sennheiser Control Cockpit)	✓	✗	Manage all mics across campus from one PC

3.4 System Reliability & Operations

Feature	Windows PC	Android TV	Notes
Sustained CPU load (1-hour @ 70%+)	✓	~ Throttles	Android SoC overheats; thermal management forced
Active cooling (fan)	✓	✗	Windows Mini PCs have fans; signage TVs are passive
Standard remote management tools	✓ RDP, SCCM, PDQ	~ Custom only	Windows has industry-standard tooling
Standard software ecosystem	✓	✗	Millions of Windows apps; signage TV is closed
BitLocker / disk encryption	✓	✗	Built-in on Windows Pro/Enterprise
Antivirus & endpoint security	✓	~	Standard enterprise tools work on Windows
Group Policy / Active Directory	✓	✗	Centralized IT policy management
Update mechanism	✓ Standard Win Update	~ Custom OTA	Windows uses standard tools; we built a custom OTA system for Android
Hardware repair / replacement	✓	~	Windows PCs are repairable; signage TV is single unit
Component-level upgrade (RAM, SSD)	✓	✗	Mini PCs are user-serviceable

3.5 Recording Workflow Features

Feature	Windows PC	Android TV	Notes
Picture-in-picture (slides + camera)	✓	~ Custom	Native multi-source compositing in our Windows application; custom-coded for Android
Multi-camera scene switching	✓	✗	Switch between front/back cameras live
Live overlay (logo, timestamp, name)	✓	✗	Built into our Windows application; impossible on Android TV
Replay buffer (last 30 sec)	✓	✗	Standard capability on Windows; rewind moments on demand
Dual output (record + live stream)	✓	~	Stream to Zoom AND record locally simultaneously
Custom scenes per class type	✓	✗	Different layouts for lectures, labs, panels
HDMI screen capture	✓ Capture card	✓ Built-in MediaProjection	Windows uses USB capture card; Android uses screen API
Audio mixing (multiple sources)	✓	~	Windows has full audio mixer; Android limited

Legend: ✓ Fully supported · ~ Partially supported / requires workarounds · ✗ Not supported

4. Lumens VC-TR1 Camera — Capability Comparison

CAMERA DEEP-DIVE

The **Lumens VC-TR1** is an AI-powered PTZ (Pan-Tilt-Zoom) camera with auto-tracking, presets, RTSP streaming, NDI broadcast, and full network management. It is a flagship-class camera designed for professional production environments — broadcast studios, telemedicine, courtrooms, classrooms.

This is a substantial investment. The question is: **how much of its capability can each platform actually use?**

4.1 Lumens VC-TR1 — Full Feature Set (As Designed)

- **AI Auto-Tracking:** Camera follows the active speaker automatically using face and voice detection.
- **PTZ Mechanism:** 12× optical zoom, 340° pan, 120° tilt. Smooth motorised movement.
- **Preset Positions:** Up to 256 saved positions, instantly recallable (e.g., "wide shot", "teacher close-up", "blackboard").
- **Dual Output:** Simultaneous USB + Ethernet RTSP — feed two systems at once.
- **Network Streaming:** RTSP, NDI 5, RTMP, SRT — all major broadcast protocols.
- **Web Management:** Browser-based admin from any computer on the network.
- **Multi-Bitrate Output:** Camera can output multiple resolution/quality streams concurrently.
- **API Control:** Full programmatic control over PTZ, presets, recording, and tracking modes.

On Windows: 100% of features unlocked

- ✓ AI auto-tracking — works directly
- ✓ PTZ control — programmable presets
- ✓ RTSP Ethernet streaming — native input to our Windows recording application
- ✓ NDI streaming — plugin available
- ✓ Web admin GUI — Chrome/Edge access
- ✓ Multi-stream output — both USB + IP
- ✓ Python/REST API control — automate everything
- ✓ Firmware updates — Lumens utility
- ✓ Centralized fleet management

Result: Camera operates as the manufacturer designed it. Investment fully realized.

On Android: ~30% of features available

- ✗ AI auto-tracking — no API path
- ✗ PTZ control — Android USB API has no PTZ
- ✗ RTSP streaming — no built-in client
- ✗ NDI streaming — Windows-only ecosystem
- ✗ Web admin GUI — no usable browser on TV
- ✗ Multi-stream — Android sees one USB stream
- ✗ API control — no path to camera's API
- ✗ Firmware updates — must remove + connect to PC
- ✓ Basic USB video capture (single stream)

Result: Camera operates as a basic USB webcam. ~70% of paid-for capability dormant.

INVESTMENT REALIZATION

The Lumens VC-TR1 has a market price reflecting its AI, PTZ, and IP capabilities. On Android, the system uses it as a plain webcam — equivalent to a ₹5,000 USB camera. On Windows, the full investment value is realized.

4.2 What "Full Automation" Actually Means

"Full automation" with the Lumens camera on Windows means scenarios like:

- **Class auto-start:** Schedule arrives, system commands camera to "Lecture Mode" preset (wide shot of room), starts recording. Zero teacher action.
- **Speaker tracking:** AI detects who's speaking, camera follows automatically. No camera operator needed.
- **Q&A handover:** When student asks question from back of room, camera auto-pans to focus on them.
- **Blackboard mode:** Teacher walks to blackboard, camera switches to "Whiteboard" preset showing the board area.
- **Multi-angle recording:** Camera outputs two streams — one for live cloud view, another for high-quality recording.

None of the above is possible on Android, because Android's USB Camera API exposes only basic video frames — no commands, no presets, no AI control surface.

> **DHUUN TEAM VALIDATION**
>
> The DHUUN team — the AV systems integration partner who supplied and configured the Lumens VC-TR1 — has confirmed that the camera's full feature set (AI auto-tracking, PTZ control, RTSP streaming, NDI output, multi-bitrate output, and centralized management) is operationally accessible on Windows-based platforms. On Android-based recording, the camera operates only in basic USB video mode; its advanced features cannot be invoked due to operating-system-level API limitations on Android.

5. Sennheiser TCC2 Microphone — Capability Comparison

AUDIO DEEP-DIVE

The **Sennheiser TeamConnect Ceiling 2 (TCC2)** is a beamforming ceiling microphone with 28 MEMS elements, dynamic beam steering, built-in DSP, Dante audio-over-IP, and centralized management. It is the gold standard for classroom and conference room audio.

5.1 Sennheiser TCC2 — Full Feature Set (As Designed)

- **TruVoiceLift Beam Steering:** Dynamically focuses microphone beam on the active speaker without manual setup.
- **28-element MEMS array:** Industry-leading sensitivity and noise rejection.
- **Dante Audio-over-IP:** Lossless digital audio over standard Ethernet, multiple streams independently routable.
- **Built-in DSP:** Acoustic echo cancellation, noise reduction, automatic gain control — all in the mic itself.
- **4 Independent Audio Channels:** Each channel can be routed to a different destination (recording, room speakers, conference call).
- **Sennheiser Control Cockpit:** Centralized PC software to manage all mics across campus from one screen.
- **API control:** Mute, level, beam settings — all controllable programmatically.
- **Health monitoring:** Real-time audio metering, SNR reporting, fault alerts.

On Windows: 100% of features unlocked

- ✓ TruVoiceLift beam steering — fully active
- ✓ Dante Virtual Soundcard — direct Dante input
- ✓ All 4 audio channels accessible
- ✓ Built-in DSP usable (AEC, NR, AGC)
- ✓ Sennheiser Control Cockpit — full management
- ✓ Real-time level metering
- ✓ API mute / unmute control
- ✓ Multi-room audio routing
- ✓ Sample-accurate sync (Dante PTP clock)

Result: Mic operates as designed. Voice clarity, beam tracking, and centralized control all active.

On Android: ~25% of features available

- ✗ Beam steering — fixed pickup pattern only
- ✗ Dante audio — Android has no Dante stack
- ✗ Multi-channel — single mono/stereo only
- ✗ DSP control — defaults locked
- ✗ Control Cockpit — no Android version
- ✗ Real-time metering — no API path
- ✗ API control — no network access from Android TV
- ✗ Multi-room routing — single endpoint
- ✓ Basic USB audio capture (analog out converted)

Result: Mic operates as a basic ceiling microphone. ~75% of paid-for capability dormant.

INVESTMENT REALIZATION

The TCC2's value is in its **beamforming** and **Dante** features. On Android, neither works. The mic operates equivalent to a ₹15,000 conference mic. On Windows, the full investment value is realized.

5.2 What This Means in Class

Scenario	On Windows	On Android
Teacher walks around the room	Beam follows automatically — clear voice everywhere	Voice level varies; far areas pick up echo
Student asks question from back row	Beam shifts to student — clear capture	Quiet, possibly inaudible in recording
Multiple speakers (panel discussion)	Beam tracks each speaker; clean audio	All voices mixed together; cross-talk
HVAC / ceiling fan noise	DSP suppresses; voice stays clear	Noise gets recorded with voice
Multi-room audio routing	Same mic feeds 3 destinations independently	Single audio stream only

DHUUN TEAM VALIDATION

The DHUUN team has confirmed that the Sennheiser TCC2's full feature set — including TruVoiceLift beamforming, Dante audio-over-IP, multi-channel audio routing, built-in DSP (echo cancellation, noise reduction, automatic gain control), and centralized management via Sennheiser Control Cockpit — is operationally accessible on Windows-based platforms. On Android-based recording, the microphone operates only in basic USB audio mode; the absence of Dante audio drivers and Sennheiser Control Cockpit support on Android prevents the advanced features from being invoked.

6. Inside the Current Android System

HOW IT WORKS TODAY

This section explains how the current Android-based system actually operates today, including the engineering work done to keep it stable. Understanding this helps clarify why the platform has the limitations it does.

6.1 System Architecture (Current)

```
┌──────────────────────────────────────────────────────────────────┐
│                  LG 55TR3DK Smart Signage TV                     │
│                                                                   │
│     ┌──────────────────────────────────────────────────────┐     │
│     │  Built-in Android Subsystem (Droidlogic SoC, 2GB RAM)  │     │
│     │                                                        │     │
│     │    ┌──────────────────────────────────────────┐        │     │
│     │    │   LectureLens Recorder App (Android APK)  │        │     │
│     │    │                                           │        │     │
│     │    │   • USB Camera (Lumens, basic mode)       │        │     │
│     │    │   • USB Audio (TCC2, basic mode)          │        │     │
│     │    │   • Screen capture (MediaProjection)      │        │     │
│     │    │   • Software encoder (CPU-based)          │        │     │
│     │    │   • Custom thermal governor               │        │     │
│     │    │   • Live streaming publisher              │        │     │
│     │    │   • OTA update client                     │        │     │
│     │    └──────────────────────────────────────────┘        │     │
│     └──────────────────────────────────────────────────────┘     │
│                                                                   │
│              [USB Port] ──────── Lumens VC-TR1                    │
│              [USB Port] ──────── Sennheiser TCC2                  │
│              [Ethernet]  ──────► Our cloud streaming + recording  │
└──────────────────────────────────────────────────────────────────┘
```

6.2 The Core Engineering Challenge

The signage TV's processor is designed for displaying digital signage content — relatively low-load tasks. Continuous recording with screen capture, camera capture, audio mixing, software encoding, and live network upload pushes the SoC well beyond its design envelope.

Within minutes of recording, the CPU heats up. Without intervention, this leads to two outcomes: either the system crashes, or the SoC silently downclocks and starts dropping frames. Neither is acceptable for classroom recording.

To keep the system stable, we engineered a custom **Thermal Governor** — software that monitors CPU usage and chip temperature in real-time, and takes protective action before the system fails.

6.3 The Thermal Governor — A 5-Tier State Machine

```
┌──────────┐   stress   ┌──────────┐   stress   ┌──────────┐   stress   ┌──────────┐
│ NORMAL   │──────────▶│ WATCH    │──────────▶│ THROTTLE │──────────▶│ COOL     │
└──────────┘            └──────────┘            └──────────┘            └──────────┘
     ▲                       ▲                       │                       │
     │                       │                       ▼                       │
     │                       │                 ┌──────────────┐              │
     │                       │                 │ Pause camera 45s │          │
     │                       │                 └──────────────┘              │
     │                       │                       │                       │
     │                       └───────────────────────┘                       │
     │                              cooling                                  │
     │                                                                       │
     └──────── recovery (CPU<80%, 90s, temp<67°C) ──────────────────────────┘
```

The system continuously monitors CPU and temperature, and transitions through five tiers based on observed stress:

Tier	Trigger	What Happens
NORMAL	CPU < 90%	Full recording — camera, screen, audio, all active.
WATCH	CPU ≥ 90% sustained for 60 seconds	Increase logging verbosity. No user-visible change. System is alerted.
THROTTLE	CPU ≥ 95% sustained for 5 minutes *OR* chip temp ≥ 70°C	**Camera is paused for 45 seconds** to let the SoC cool. Recording continues with screen + audio only. Camera then auto-resumes.
COOL	CPU ≥ 98% sustained for 2 minutes *OR* chip temp ≥ 78°C (emergency)	Extended camera pause (~90 seconds) to give SoC more cooling time. Recording continues without camera.
ABORT	Critical hardware fault	Recording stops gracefully, MP4 finalized, alert sent to admin.

6.4 The Cycling Camera Behavior (v3.7.3 — Current)

In real classroom use, the system frequently enters THROTTLE state during 1-hour classes because of sustained CPU pressure. Earlier versions of our software would drop the camera permanently in this case (resulting in an audio-only recording for the rest of class).

In the current version (v3.7.3), we engineered a **cycling camera behavior** that maximizes camera-on time:

```
Class starts                                Class ends
|                                                |
▼                                                ▼

NORMAL ──► THROTTLE ──► (45s pause) ──► NORMAL ──► THROTTLE ──► (45s pause) ──► NORMAL ──► ...
 ▲          |                            ▲          |
 |          ▼                            |          ▼
 |       Camera OFF                      |       Camera OFF
 |       Audio + Screen                  |       Audio + Screen
 |       continue recording              |       continue recording
 |                                       |
 └── Camera back ON when CPU < 80% ──────┘
     for 90 seconds and temp < 67°C
     (whichever condition met first)
```

This cycling logic means: instead of losing camera permanently after the first overheat event, the camera comes back as soon as the SoC cools enough. In practice, this gives the client substantially more camera footage during long classes.

> **HOWEVER — A FUNDAMENTAL LIMIT REMAINS**
>
> Even with the cycling camera behavior, there are still 45-second to 90-second windows during a 1-hour class where the camera feed is dropped. Audio and screen capture continue, but the visual record has gaps. **This is a hardware limitation that cannot be fully eliminated by software** — the SoC is simply not powerful enough to sustain full recording load for 60 minutes without thermal management.

6.5 What the Thermal Governor Does NOT Solve

- **Hardware encoder limitations:** The TV's video encoder is shared with display rendering. Even at idle, encoding 1080p continuously consumes most of its capacity.
- **USB bandwidth bottleneck:** USB 2.0 ports limit camera+mic bandwidth; no way to expand.
- **Inability to use camera/mic IP features:** Android USB stack does not expose RTSP, Dante, PTZ, or DSP features.
- **Storage I/O slowness:** eMMC storage is ~5× slower than NVMe SSD; impacts buffer flushes during peak load.
- **Memory pressure:** 2GB RAM is tight when running encoder + screen capture + live streaming publisher + UI.
- **Update mechanism complexity:** Required us to build a custom OTA system, which adds engineering maintenance overhead.

7. Architecture Transition — Current to Proposed

SYSTEM ARCHITECTURE

This section illustrates how the system structure transitions from the current Android-based architecture to the proposed Windows-based architecture, and how the Windows architecture enables reliable concurrent multi-room recording at scale.

7.1 Current Architecture (Android-Based)

```
┌──────────────────────────────────────────────────────────────────┐
│                  LG 55TR3DK Smart Signage TV                     │
│                                                                   │
│   USB ◄── Camera (Lumens VC-TR1) — basic USB mode only           │
│   USB ◄── Mic (Sennheiser TCC2) — basic USB audio mode only      │
│                                                                   │
│   ▼                                                               │
│   Built-in Android SoC                                            │
│     • Software encoder (CPU-based)                                │
│     • Custom thermal management                                   │
│     • Live stream publisher                                       │
│   ▼                                                               │
│   [Ethernet] ──► Our cloud streaming + recording infrastructure  │
└──────────────────────────────────────────────────────────────────┘

Recording path:  Camera → TV SoC → Encoder → Live Stream → Cloud
                                                            Recording
Display path:    TV's Android UI on the same TV screen.
```

Bottlenecks observed in current architecture

- The TV SoC handles capture, encode, and upload simultaneously — the device overheats during long recordings
- Camera and microphone are limited to USB basic modes; their advanced features (RTSP, Dante, PTZ, beamforming) are not reachable from Android
- The cloud recording component is shared across all rooms — concurrent recording capacity must be scaled centrally
- If the TV reboots or the Android app crashes, the recording pipeline must restart entirely
- Recording quality is directly tied to network reliability at the moment of class

7.2 Proposed Architecture (Windows-Based)

```
┌──────────────────────────────────────────────────────────────────┐
│                          Windows Mini PC                         │
│                                                                  │
│   Ethernet ◄── Camera (Lumens VC-TR1) — RTSP, full features      │
│   Ethernet ◄── Mic (Sennheiser TCC2)  — Dante, full features     │
│                                                                  │
│   ▼                                                              │
│   Our Windows Recording Application (D&R AI Solutions)           │
│     • Hardware video encoder (Intel Quick Sync)                  │
│     • Multi-source compositing (camera + screen + audio)         │
│     • Local recording to SSD — always-on, network-independent    │
│     • Live streaming via our video conferencing platform         │
│     • Background cloud upload — queued, resumable                │
│   ▼                                                              │
│   ┌──────────────────────┐     ┌──────────────────────────────┐  │
│   │ Local SSD recording  │     │ Live stream → our cloud      │  │
│   │ (primary, immediate) │     │ (only when remote view used) │  │
│   └──────────────────────┘     └──────────────────────────────┘  │
│              │                                                   │
│              ▼ background upload                                 │
│   [Ethernet] ──► Our cloud archive                               │
│                                                                  │
│   Display ──HDMI──► LG TV (used purely as a monitor;             │
│                            recording does not depend on it)      │
└──────────────────────────────────────────────────────────────────┘

Recording path:  Camera → Mini PC → Hardware encoder → Local SSD ─┐
                                                                  │
                                                Background upload ─▼
                                                            Cloud Archive
```

Why this architecture works better

- Recording happens on dedicated computing hardware, not on display hardware
- Camera and mic use their full-featured Ethernet/Dante paths instead of basic USB
- Local SSD recording is the source of truth — network interruption does not affect the recording itself
- Each Mini PC operates independently — no shared real-time bottleneck across rooms
- Live streaming becomes optional — used only when a remote viewer needs to watch the class live

7.3 Multi-Room Concurrent Recording — Side-by-Side

Current — Concurrent Recording on Android

```
Room 1 TV ─┐
Room 2 TV ─┼─► Shared cloud recording
Room 3 TV ─┤   ◄── Capacity must scale
Room 4 TV ─┘       centrally with rooms

Each TV depends on the same
cloud recording capacity to be
available at the same time.
```

In our pilot testing, this shared cloud component was observed to constrain concurrent recordings. Adding more rooms requires central infrastructure scaling.

Proposed — Concurrent Recording on Windows

```
Room 1 PC ──► local SSD ──► upload
Room 2 PC ──► local SSD ──► upload
Room 3 PC ──► local SSD ──► upload
Room 4 PC ──► local SSD ──► upload

Each Mini PC records independently.
Cloud upload is asynchronous — no
shared real-time bottleneck.
```

Each room is autonomous. Recording continues even if cloud is briefly unreachable. Adding rooms = adding Mini PCs; no central infrastructure changes required.

MULTI-ROOM CAPACITY

With the Windows architecture, recording capacity scales linearly with the number of rooms — each Mini PC contributes its own complete recording capability. There is no shared real-time recording bottleneck. The cloud only handles background upload of completed recordings, which is an asynchronous batch operation that scales straightforwardly.

7.4 Recording Continuity — Failure Mode Comparison

Scenario	Windows Architecture	Android Architecture
Internet drops during class	Local recording continues uninterrupted; cloud upload queues	Live streaming pipeline impacted; recording quality affected
Display TV powers off / disconnected	Recording continues — Mini PC is independent of the display	Recording stops — TV is the recorder
Power outage	UPS bridges short outages; auto-resume on power return	TV restart resets pipeline; manual recovery often needed
Application crash	Service auto-restart; recording resumes within seconds	Custom recovery; recording often restarts from beginning
Camera connection issue	Ethernet RTSP path is more stable; auto-reconnect	USB camera disconnect events observed in pilot
5+ rooms recording simultaneously	Each room records independently — no contention	Shared cloud capacity must be sufficient for all concurrent rooms

8. Smart Notebook & Teaching Tools

TEACHER CONTINUITY

A reasonable concern when changing the platform is whether teachers will face disruption to their existing workflow. Smart notebook applications, annotation tools, slide presentation, and interactive whiteboard features must continue to be available — and ideally, become richer.

Both Android and Windows platforms support smart notebook applications. Windows, however, has a substantially larger and more mature ecosystem in this category, particularly in the education segment.

8.1 Smart Notebook Application Availability

Android Smart Notebook Apps

- Apps available through the Android app ecosystem, subject to the signage TV's app installation policy
- Touch interaction depends on the TV's touch overlay (where present)
- Annotation, drawing, and basic save/share are supported by mainstream apps
- Cloud sync via Google or app-specific backends
- Stylus pressure support is limited — most signage TVs are not pen-enabled by design

Functional, but constrained by signage TV's app ecosystem and input hardware.

Windows Smart Notebook Apps

- **Microsoft Whiteboard** — free, cloud-synced, multi-user collaboration
- **Microsoft OneNote** — full stylus, ink-to-math, OCR, Office integration
- **SMART Notebook** — long-established standard in education classrooms
- **Promethean ActivInspire** — established interactive whiteboard software
- **Open-source options** — OpenBoard, Open-Sankoré (free, classroom-focused)
- **Web-based options** — Jamboard, Miro, Mural, FigJam (browser, no install)
- Native Windows Ink API — hardware-accelerated drawing, palm rejection
- Stylus support across compatible touch displays with pressure sensitivity

Mature, broad ecosystem with category-defining tools used across global education.

8.2 Feature Parity Map

Capability	Windows Smart Notebook	Android Smart Notebook
Free-hand annotation / drawing	✓ All notebook apps	✓
Stylus / pen with pressure sensitivity	✓ Native Windows Ink	~ Hardware-dependent
Save and reload notebook	✓	✓
Cloud sync across devices	✓ OneDrive, Google Drive, etc.	✓
Multi-user live collaboration	✓ Whiteboard, Jamboard	~
Image / PDF import as background	✓	✓
Mathematical equation input	✓ OneNote ink-to-math	~
Notebook content captured in lecture recording	✓ Automatic via screen capture	~ Custom-coded
Hand-writing to text recognition (OCR)	✓ OneNote, Whiteboard	~
Share notebook to students	✓ Multiple methods	✓

8.3 Teacher Onboarding

For teachers transitioning from Android-based notebook apps to Windows-based notebook apps, the user experience is straightforward:

- **Drawing and annotation interactions are visually consistent** across both platforms — pinch to zoom, drag to pan, tap to write.
- **Windows notebook apps are widely used in education globally** — teachers may already be familiar with these from previous institutions, training programs, or personal use.
- **Stylus and touch input on Windows are mature** — Windows Ink has been a core operating-system feature since 2016 and is the standard input model for educational tablets and pen displays.
- **Notebook content integrates naturally with the lecture recording** — anything drawn on the notebook becomes part of the recorded lecture without separate handling.

8.4 Other Teaching Tools — Continuity Across Categories

Teaching Tool Category	Windows	Android Signage TV
Slide presentation (PowerPoint, Keynote, Slides)	✓ Native PowerPoint, web for others	✓ Web-based
PDF viewer with annotation	✓ Many options including free	✓
Web browsing (Chrome, Edge, Firefox)	✓ All major browsers	~ Limited browser on signage TV
Video conferencing (Zoom, Meet, Teams)	✓ Native Windows clients	~ Web or limited native
Video playback (YouTube, embedded videos)	✓	✓
Document camera integration	✓ USB or HDMI	~ Limited
Educational software (Mathematica, GeoGebra, etc.)	✓ Native Windows apps	✗ Most not on Android signage
LMS integration (Moodle, Canvas, etc.)	✓ Browser + native apps	~ Browser only

TEACHER WORKFLOW CONTINUITY

Every teaching workflow currently performed on the Android signage TV has a direct, mature equivalent on Windows — typically with richer feature support and broader software choice. Teachers will not lose access to any current capability; in many cases, they gain new options.

9. What More Can Be Done on Android

ANDROID OPTIMIZATION

If the decision is to continue with Android, there are still optimizations possible. Each has trade-offs.

9.1 Software-Side Optimizations

Optimization	Benefit	Trade-off
Lower video bitrate (3 Mbps → 2 Mbps)	Reduces encoder CPU load by ~25%	Lower video quality, especially noticeable for fast motion
Drop screen capture when CPU rises	~30% CPU saving when in throttle window	Slides/whiteboard not in recording during throttle
Lower frame rate (30fps → 20fps)	~33% encoder load reduction	Choppy video — less acceptable for client
Skip in-class composition (post-process later)	Records raw streams; cloud composites	Higher cloud cost; live preview not possible
Cap class duration (60 min → 45 min)	Avoids worst thermal phase	Pedagogical limit; not always practical

9.2 Hardware-Side Optimizations

Optimization	Benefit	Trade-off
External USB cooling fan (₹500/room)	Drops SoC temp 5–8°C, reduces throttle frequency	Noise, cabling, power; not all rooms allow
Powered USB 3.0 hub	More stable USB camera/mic supply	Doesn't help with encoder/CPU bottleneck
Replace TV with newer signage model	~20% better SoC, slightly more headroom	₹85,000 capex per room; same architecture
External Android device (NVIDIA Shield, ~₹35K)	Significantly better SoC than signage TV	Still Android — same API limits with USB

HONEST ENGINEERING ASSESSMENT

The optimizations above can extend the current system's useful life and improve reliability marginally. None of them, however, can unlock the Lumens or TCC2 hardware features that require RTSP, Dante, or PTZ control — those capabilities are blocked at the Android operating system level, not by our software. Squeezing more performance from Android improves the same plateau; it doesn't move us to a new one.

9.3 Could the Current System Run More Smoothly at 720p?

A practical question is whether reducing the recording quality from 1080p to 720p would allow the existing Android-based system to operate more reliably for 1-hour classes. This is a meaningful question because lower resolution means lower encoding workload, and lower encoding workload means less thermal stress on the TV's SoC.

Engineering Analysis — 1080p vs 720p on the Existing System

Metric	At 1080p (current)	At 720p	Net Effect
Pixels per frame	2.07 million (1920×1080)	0.92 million (1280×720)	~55% fewer pixels
Video bitrate (H.264)	~3 Mbps	~1.5 Mbps	~50% lower
Software encoder CPU load	~80–90% sustained	~50–60% sustained	~30–40% reduction
Thermal throttle events per 1-hour class	2–4 events	0–1 events (estimated)	Significant improvement
Storage per 1-hour recording	~1.4 GB	~700 MB	~50% less
Cloud upload time	Baseline	~50% faster	Faster turnaround
Whiteboard / blackboard text legibility	Good	Reduced	Visible degradation
Slide detail clarity (small fonts, diagrams)	Good	Reduced	Small text harder to read
Far-room student face detail	Recognizable	Less detailed	Reduced clarity

What 720p Would Improve

- 1-hour classes would likely complete with significantly fewer or zero camera-drop cycles
- Lower SoC heat — thermal throttle state reached less often, if at all
- Faster cloud upload, less network bandwidth consumed
- Less storage pressure on the TV's eMMC (which is slow)
- Reduced risk of memory pressure during long classes

What 720p Does NOT Solve

- **Lumens VC-TR1 advanced features remain unreachable** — AI auto-tracking, PTZ control, RTSP, NDI, multi-bitrate streaming. These are blocked at the Android operating system level, not by quality settings.
- **Sennheiser TCC2 advanced features remain unreachable** — beamforming, Dante, multi-channel routing, DSP control. Same reason.
- **Multi-room concurrent capacity** still constrained by shared cloud recording.
- **2-hour+ continuous classes** still thermally challenging — the SoC is fundamentally not designed for sustained recording loads of this length.
- **Custom OTA system, custom thermal management, custom recovery logic** all still required.

> **ENGINEERING VERDICT ON 720P**
>
> Reducing recording quality to 720p would meaningfully improve thermal stability for 1-hour classes, with fewer or zero camera-drop cycles per class. However, this comes at the cost of recording detail — particularly for board-heavy and slide-heavy lectures where small text becomes harder to read on playback. Crucially, 720p does not unlock any of the Lumens or Sennheiser advanced features, and does not address platform-level limitations such as multi-room concurrent capacity or 2-hour class viability.

9.4 What Stays Permanently Unavailable on Android

Regardless of any optimization, the following will remain unavailable on Android-based recording:

- Lumens AI auto-tracking
- Lumens PTZ control and presets
- Lumens RTSP / NDI / multi-stream output

- TCC2 beam steering
- TCC2 Dante audio-over-IP
- TCC2 multi-channel routing
- TCC2 centralized control via Sennheiser Control Cockpit
- Standard remote management tools (RDP, Group Policy, SCCM)
- 4K recording capability
- 2-hour+ continuous recording without thermal cycling

10. What Is Uniquely Possible on Windows

WINDOWS CAPABILITIES

Windows is not just "a different platform" — it is the platform that the Lumens and TCC2 hardware was designed for. Here are capabilities that become available on Day 1 of a Windows deployment.

10.1 Day-1 Out-of-the-Box Capabilities

★ HEADLINING FEATURE: FULL CAMERA/MIC AUTOMATION

Windows can speak directly to the Lumens camera (RTSP, API) and TCC2 mic (Dante, API). The system can automatically position the camera based on class type, follow the speaker via AI, switch to blackboard view, and steer microphone beams — without any manual intervention. None of this requires custom development; it uses standard manufacturer integration.

★ RECORDING INDEPENDENCE (DECOUPLED ARCHITECTURE)

On Windows, recording happens on the Mini PC. The display is a separate concern. Even if the display fails, the user logs out, or the screen is unplugged, recording continues. Power outage with UPS — recording resumes without data loss. This is the architectural pattern used by professional broadcast studios.

★ MULTI-CAMERA SUPPORT

Add a second camera (front of room + back of room, or main + document camera) at any time. Our Windows recording application supports multi-camera scene switching natively. Currently impossible on Android.

★ CONCURRENT STREAM OUTPUT

Same recording can simultaneously: (a) save to local disk, (b) upload to cloud, (c) live-stream to Zoom/Meet/YouTube. All from the same Mini PC, in parallel. No additional infrastructure.

★ STANDARD IT OPERATIONS

No more custom OTA system to maintain. Windows Update, Group Policy, Active Directory, BitLocker, antivirus, RDP — all the tools your IT team already knows. Onboarding new IT staff is trivial; signage TV management is a niche skill.

10.2 Future Capabilities Easily Added Later

Capability	Effort to Add (Windows)	Possible on Android?
Live captions / transcription	Plugin install	✗
AI summarization (post-class)	Cloud integration	~ Cloud only
Auto-attendance via face recognition	Camera API	✗
Real-time translation overlay	Software	✗
Dolby Vision / HDR recording	Encoder profile	✗
4K recording	Encoder profile	✗
3rd-party LMS integration	Standard SDKs	~
Backup recording to NAS	SMB built-in	✗

11. Summary

FINAL VIEW

11.1 Platform Comparison at a Glance

Dimension	Windows PC	Android TV (Current)
Lumens VC-TR1 features used	100% (full automation)	~30% (basic webcam mode)
Sennheiser TCC2 features used	100% (Dante + beamforming)	~25% (basic mic)
1-hour recording reliability	~99.5%	~92% (with thermal cycling)
Camera dropout during class	0 events	2–4 cycles per hour (45–90s each)
2-hour+ class support	Yes	Significant degradation
4K recording	Yes	No
Future feature additions	Standard plugins	Often blocked by OS limits
Hardware repair / upgrade	Component-level	Replace whole TV
Custom software maintenance	Minimal (standard tools)	Significant (custom OTA, custom thermal logic)
Recording architecture	Decoupled from display	Coupled to TV

11.2 The Investment Realization Argument

The institution has invested in professional-grade AV equipment — the Lumens VC-TR1 and Sennheiser TCC2. These are flagship-class products with capabilities the institution paid for. The platform decision determines how much of that investment is actually realized:

~30%
CAMERA + MIC CAPABILITY
REALIZED ON ANDROID

100%
CAMERA + MIC CAPABILITY
REALIZED ON WINDOWS

11.3 Why This Matters Going Forward

The platform choice is not just about "Does it record today?" Both platforms record today. The choice is about:

- **How much of your AV equipment investment is actually used.**
- **What capabilities you can add as the system evolves.**
- **How much custom software the engineering team must maintain.**
- **How recordings hold up during long classes, peak load, edge conditions.**
- **How easy or hard it becomes to scale to 50, 200, 500 classrooms.**

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This document represents a feature-wise capability analysis based on hardware specifications, pilot testing across multiple classrooms, and direct engineering observation of both platforms.

For technical clarification or follow-up discussion, please contact the D&R team.